INCH-POUND

MIL-PRF-19500/746
15 May 2008

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, FIELD EFFECT RADIATION HARDENED
(TOTAL DOSE AND SINGLE EVENT EFFECTS) TRANSISTOR, N-CHANNEL,
SILICON, TYPES 2N7587U3, 2N7587U3C, 2N7589U3, 2N7589U3C, 2N7591U3,
2N7591U3C, 2N7593U3, AND 2N7593U3C,
JANTXVR, JANTXVF, JANSR, AND JANSF

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of
this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for a N-channel, enhancement-mode, MOSFET, radiation hardened (total dose and single event effects (SEE)), power transistor. Two levels of product assurance are provided for each device type as specified in MIL-PRF-19500, with avalanche energy maximum rating (E_{AS}) and maximum avalanche current (I_{AS}).

1.2 Physical dimensions. See figure 1, (surface mount, TO-276AA, SMD-0.5) with metal lid (U3 suffix) or ceramic lid (U3C suffix).

1.3 Maximum ratings. T_A = +25°C, unless otherwise specified.

Type	P_T (1) T_C = +25°C	P_T T_A = +25°C	$R_{\theta JC}$ (2)	V_{DS}	V_{DG}	V_{GS}	I_{D1} (3) (4) T_C =+25°C	I_{D2} T_C = +100°C	I_S	I_{DM} (5)	T_J and T_{STG}	V_{ISO} 70,000 ft. altitude
	W	W	°C/W	V dc	V dc	V dc	A dc	A dc	A dc	A (pk)	°C	V dc
2N7587U3, 2N7587U3C	75	1.56	1.67	100	100	±20	22	19	22	88		
2N7589U3, 2N7589U3C	75	1.56	1.67	150	150	±20	19	12	19	76	-55 to +150	
2N7591U3, 2N7591U3C	75	1.56	1.67	200	200	±20	16	10	16	64		
2N7593U3, 2N7593U3C	75	1.56	1.67	250	250	±20	12.4	7.8	12.4	49.6		250

(1) Derate linearly by 0.6 W/°C for T_C > +25°C.
(2) See figure 2, thermal impedance curves.
(3) The following formula derives the maximum theoretical I_D limit. I_D is limited by package and internal wires and may be limited by pin diameter:

$$I_D = \sqrt{\frac{T_{JM} - T_C}{(R_{\theta JC}) \times (R_{DS}(on) \text{ at } T_{JM})}}$$

(4) See figure 3, maximum drain current graph.
(5) I_{DM} = 4 X I_{D1}; I_{D1} as calculated by footnote (3).

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1.4 Primary electrical characteristics at T_C = +25°C.

Type	Min $V_{(BR)DSS}$ V_{GS} = 0 I_D = 1.0mA dc	$V_{GS(TH)1}$ $V_{DS} \geq V_{GS}$ I_D = 1.0 mA dc		Max I_{DSS1} V_{GS} = 0 V_{DS} = 80% of rated V_{DS}	Max $r_{DS(on)}$ (1) V_{GS} = 12V, $I_D = I_{D2}$		E_{AS}
					T_J = +25°C	T_J = +150°C	
	V dc	V dc		µA dc	Ω	Ω	mJ
		Min	Max				
2N7587U3, 2N7587U3C	100	2.0	4.0	10	0.042	0.084	73
2N7589U3, 2N7589U3C	150	2.0	4.0	10	0.088	0.207	60
2N7591U3, 2N7591U3C	200	2.0	4.0	10	0.130	0.300	60
2N7593U3, 2N7593U3C	250	2.0	4.0	10	0.210	0.494	56

(1) Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

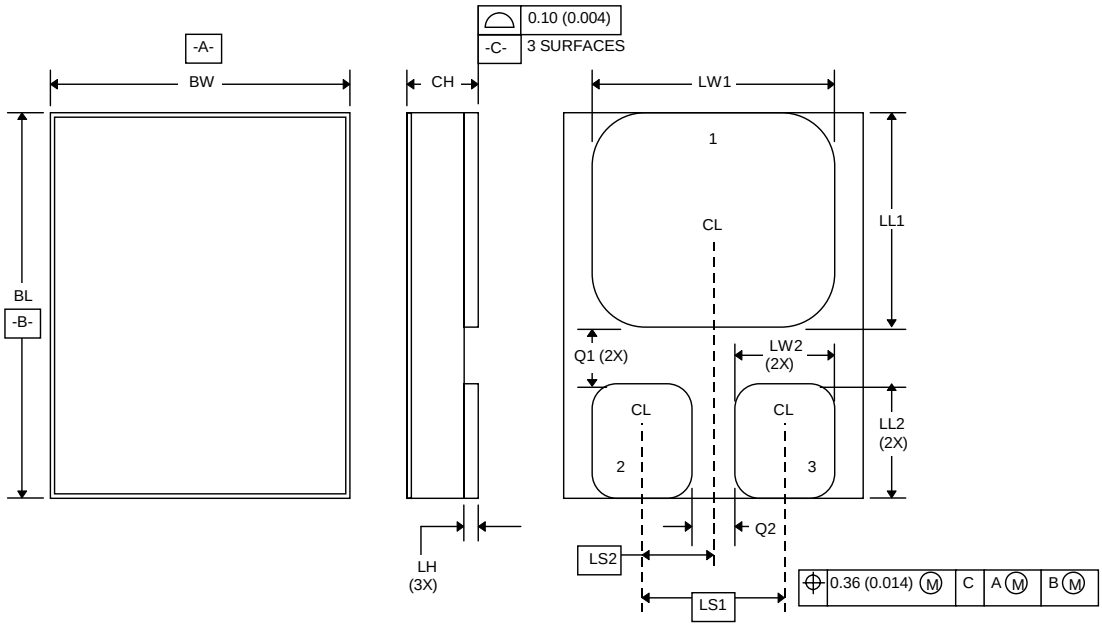
DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.395	.405	10.04	10.28
BW	.291	.301	7.40	7.64
CH (for U3)		.124		3.15
CH (for U3C)		.1335		3.39
LH	.010	.020	0.25	0.51
LW1	.281	.291	7.14	7.39
LW2	.090	.100	2.29	2.54
LL1	.220	.230	5.59	5.84
LL2	.115	.125	2.93	3.17
LS1	.150 BSC		3.81 BSC	
LS2	.075 BSC		1.91 BSC	
Q1	.030		0.762	
Q2	.030		0.762	
TERM 1	Drain			
TERM 2	Gate			
TERM 3	Source			

NOTES:
1. Dimension are in inches.
2. Millimeters are given for information only.
3. The lid shall be electrically isolated from the drain, gate, and source.
4. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.
5. Metal lid: U3 suffix; Ceramic lid: U3C suffix.

* FIGURE 1. Dimensions and configuration (TO-276AA, SMD-0.5), with metal lid or ceramic lid).

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3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturer's list (QML) before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1 (TO-276AA, SMD-0.5, with metal lid or ceramic lid) herein.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Multiple chip construction. Multiple chip construction is not permitted to meet the requirements of this specification.

3.5 Electrostatic discharge (ESD) protection. The devices covered by this specification require electrostatic discharge protection (see 3.5.1).

3.5.1 Handling. Metal oxide semiconductor (MOS) devices must be handled with certain precautions to avoid damage due to the accumulation of static charge. However, the following handling practices are recommended (see 3.5).

a. Devices should be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber or silk in MOS areas.

f. Maintain relative humidity above 50 percent if practical.

g. Care should be exercised during test and troubleshooting to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source, $R \leq$ or 100 kΩ, whenever bias voltage is applied drain to source.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.9 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500. Alternate flow is allowed for qualification inspection in accordance with figure 4 of MIL-PRF-19500.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

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4.3 Screening (JANS and JANTXV). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table E-IV of MIL-PRF-19500) (1) (2)	Measurement	
	JANS	JANTXV
(3)	Gate stress test (see 4.3.1)	Gate stress test (see 4.3.1)
(3)	Method 3470 of MIL-STD-750, E_{AS} (see 4.3.2)	Method 3470 of MIL-STD-750, E_{AS} (see 4.3.2)
(3) 3c	Method 3161 of MIL-STD-750, thermal impedance, (see 4.3.3)	Method 3161 of MIL-STD-750, thermal impedance, (see 4.3.3)
9	Subgroup 2 of table I herein I_{GSSF1}, I_{GSSR1}, I_{DSS1} as a minimum	Not applicable
10	Method 1042 of MIL-STD-750, test condition B	Method 1042 of MIL-STD-750, test condition B
11	I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $r_{DS(ON)1}$, $V_{GS(TH)1}$ Subgroup 2 of table I herein. ΔI_{GSSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±10 μA dc or ±100 percent of initial value, whichever is greater.	I_{GSSF1}, I_{GSSR1}, I_{DSS1}, $r_{DS(ON)1}$, $V_{GS(TH)1}$ Subgroup 2 of table I herein.
12	Method 1042 of MIL-STD-750, test condition A	Method 1042 of MIL-STD-750, test condition A
13	Subgroups 2 and 3 of table I herein ΔI_{GSSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±10 μA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(ON)1}$ = ±20 percent of initial value. $\Delta V_{GS(TH)1}$ = ±20 percent of initial value.	Subgroups 2 and 3 of table I herein ΔI_{GSSF1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{GSSR1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±10 μA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(ON)1}$ = ±20 percent of initial value. $\Delta V_{GS(TH)1}$ = ±20 percent of initial value.

(1) At the end of the test program, I_{GSSF1}, I_{GSSR1}, and I_{DSS1} are measured.
(2) An out-of-family program to characterize I_{GSSF1}, I_{GSSR1}, I_{DSS1} and $V_{GS(th)1}$ shall be invoked.
(3) Shall be performed anytime after temperature cycling, screen 3a; and does not need to be repeated in screening requirements.

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4.3.1 Gate stress test. Apply V_{GS} = 24 V minimum for t = 250 μs minimum.

4.3.2 Single pulse avalanche energy (E_{AS}).

a. Peak current ..... $I_{AS} = I_{D1}$.

b. Inductance:..... $\left[\frac{2E_{AS}}{(I_{D1})^2}\right]\left[\frac{V_{BR} - V_{DD}}{V_{BR}}\right]$ mH minimum.

c. Gate to source resistor (R_{GS})..... $25 \leq R_{GS} \leq 200\ \Omega$.

d. Supply voltage (V_{DD})..... V_{DD} = 25 V dc, except V_{DD} = 50 V dc (2N7593), up to rated V_{DS}.

e. Peak gate voltage (V_{GS}) ..... 12 V, up to maximum rated V_{GS}.

f. Initial case temperature ..... T_C = +25°C +10°C, -5°C.

g. Number of pulses to be applied..... 1 pulse minimum.

4.3.3 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3161 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{SW}, (and V_H where appropriate). Measurement delay time (t_{MD}) = 30 - 60 μs max. See table III, group E, subgroup 4 herein.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500. Alternate flow is allowed for conformance inspection in accordance with figure 4 of MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500 and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIA (JANS) and table E-VIB (JANTXV) of MIL-PRF-19500, and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.4.2.1 Group B inspection, table E-VIA (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B3	1051	Test condition G, 100 cycles.
B3	2077	Scanning electron microscope (SEM).
B4	1042	Intermittent operation life, condition D, 2,000 cycles; t_{on} = 30 seconds minimum.
B5	1042	Accelerated steady-state gate bias, condition B, V_{GS} = rated; T_A = +175°C, t = 24 hours minimum; or T_A = +150°C, t = 48 hours minimum.
B5	1042	Accelerated steady-state reverse bias, condition A, V_{DS} = rated; T_A = +175°C, t = 120 hours minimum; or T_A = +150°C, t = 240 hours minimum.
B5	2037	Test condition D.

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4.4.2.2 Group B inspection, table E-VIB (JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
B2	1051	Test condition G, 25 cycles.
B3	1042	Intermittent operation life, condition D, 2,000 cycles; t_{on} = 30 seconds minimum.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500 and as follows. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

Subgroup	Method	Condition
C2	2036	Terminal strength is not applicable.
C5	3161	See 4.3.3, $R_{\theta JC}$ = 1.67 °C/W.
C6	1042	Intermittent operation life, condition D, 6,000 cycles; t_{on} = 30 seconds minimum.

4.4.4 Group D inspection. Group D inspection shall be conducted in accordance with table E-VIII of MIL-PRF-19500 and table II herein.

4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table III herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.4.5.1 SEE. Design capability shall be tested on the initial qualification and thereafter whenever a major die design or process change is introduced. See the safe operation area graph herein. Electrical measurements (end-points) shall be in accordance with table III herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Condition		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Thermal impedance 2/	3161	See 4.3.3	$Z_{\theta JC}$			°C/W
Breakdown voltage drain to source	3407	Bias condition C, V_{GS} = 0 V, I_D = 1 mA dc	$V_{(BR)DSS}$			
2N7587U3, 2N7587U3C				100		V dc
2N7589U3, 2N7589U3C				150		V dc
2N7591U3, 2N7591U3C				200		V dc
2N7593U3, 2N7593U3C				250		V dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = 1 mA dc	$V_{GS(TH)1}$	2.0	4.0	V dc
Gate current	3411	V_{GS} = +20 V dc, bias condition C, V_{DS} = 0 V	I_{GSSF1}		+100	nA dc
Gate current	3411	V_{GS} = -20 V dc, bias condition C, V_{DS} = 0 V	I_{GSSR1}		-100	nA dc
Drain current	3413	V_{GS} = 0 V dc, bias condition C, V_{DS} = 80 percent of rated V_{DS},	I_{DSS1}		10	µA dc
Static drain to source on-state resistance	3421	V_{GS} = 12 V dc, condition A, pulsed (see 4.5.1), $I_D = I_{D2}$	$r_{DS(ON)1}$			
2N7587U3, 2N7587U3C					0.042	Ω
2N7589U3, 2N7589U3C					0.088	Ω
2N7591U3, 2N7591U3C					0.130	Ω
2N7593U3, 2N7593U3C					0.210	Ω
Forward voltage	4011	V_{GS} = 0 V dc, condition A, pulsed (see 4.5.1), $I_D = I_{D1}$	V_{SD}		1.2	V dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Condition		Min	Max	
Subgroup 3						
High temperature operation		$T_C = T_J = +125°C$				
Gate current	3411	$V_{GS} = \pm 20$ V dc, bias condition C, $V_{DS} = 0$ V	I_{GSS2}		±200	nA dc
Drain current	3413	$V_{GS} = 0$ V dc, bias condition C, V_{DS} = 80 percent of rated V_{DS}	I_{DSS2}		25	µA dc
Static drain to source on-state resistance	3421	V_{GS} = 12 V dc, condition A, pulsed (see 4.5.1), $I_D = I_{D2}$	$r_{DS(ON)3}$			
2N7587U3, 2N7587U3C					0.080	Ω
2N7589U3, 2N7589U3C					0.176	Ω
2N7591U3, 2N7591U3C					0.273	Ω
2N7593U3, 2N7593U3C					0.441	Ω
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = 1 mA dc	$V_{GS(TH)2}$	1.0		V dc
Low temperature operation		$T_C = T_J = -55°C$				
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS(TH)3}$, I_D = 1 mA dc	$V_{GS(TH)3}$		5.0	V dc
Subgroup 4						
Forward transconductance	3475	$I_D = I_{D2}$, V_{DD} = 15 V dc (see 4.5.1)	g_{FS}			
2N7587U3, 2N7587U3C				14		S
2N7589U3, 2N7589U3C				13		S
2N7591U3, 2N7591U3C				10		S
2N7593U3, 2N7593U3C				8.8		S
Gate series resistance	3402	Resonant frequency	R_G	1	2	Ω
Subgroup 5						
Safe operating area test	3474	See figure 4; tp = 10 ms min. VDS = 80 percent of max. rated VDS				
Electrical measurements		See table I, subgroup 2				
Subgroup 6						
Not applicable						

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Condition		Min	Max	
Subgroup 7						
Gate charge	3471	Condition B, $I_D = I_{D1}$, V_{GS} = 12 V dc V_{DD} = 50 percent of rated V_{DS}				
On-state gate charge and turn-off gate charge			$Q_{G(ON)}$ $Q_{G(OFF)}$			
2N7587U3, 2N7587U3C					35	nC
2N7589U3, 2N7589U3C					50	nC
2N7591U3, 2N7591U3C					42	nC
2N7593U3, 2N7593U3C					40	nC
Gate to source charge (turn-on and turn-off)			Q_{GS1} Q_{GS2}			
2N7587U3, 2N7587U3C					13	nC
2N7589U3, 2N7589U3C					15	nC
2N7591U3, 2N7591U3C					15	nC
2N7593U3, 2N7593U3C					14	nC
Gate to drain charge (turn-on and turn-off)			Q_{GD1} Q_{GD2}			
2N7587U3, 2N7587U3C					14	nC
2N7589U3, 2N7589U3C					18	nC
2N7591U3, 2N7591U3C					15	nC
2N7593U3, 2N7593U3C					11	nC
Reverse recovery time	3473	di/dt = -100 A/µs, $V_{DD} \leq$ 50 V $I_D = I_{D1}$	t_{rr}			
2N7587U3, 2N7587U3C					355	ns
2N7589U3, 2N7589U3C					300	ns
2N7591U3, 2N7591U3C					346	ns
2N7593U3, 2N7593U3C					654	ns

1/ For sampling plan, see MIL-PRF-19500.
2/ This test required for the following end-point measurements only:
Group B, subgroups 2 and 3 (JANTXV).
Group B, subgroups 3 and 4 (JANS).
Group C, subgroup 2 and 6.
Group E, subgroup 1.

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TABLE II. Group D inspection.

Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Pre-irradiation limits		Post-irradiation limits		Unit
				R and F		R and F		
	Method	Conditions		Min	Max	Min	Max	
Subgroup 1								
Not applicable								
Subgroup 2		T_C = + 25°C						
Steady-state total dose irradiation (V_{GS} bias) 4/	1019	V_{GS} = 12 V; V_{DS} = 0						
Steady-state total dose irradiation (V_{DS} bias) 4/	1019	V_{GS} = 0; V_{DS} = 80 percent of rated V_{DS}(preirradiation)						
End-point electricals:								
Breakdown voltage, drain to source	3407	Bias condition C, V_{GS} = 0; I_D = 1 mA	$V_{(BR)DSS}$					
2N7587U3, 2N7587U3C				100		100		V dc
2N7589U3, 2N7589U3C				150		150		V dc
2N7591U3, 2N7591U3C				200		200		V dc
2N7593U3, 2N7593U3C				250		250		V dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$ I_D = 1 mA	$V_{GS(th)1}$	2.0	4.0	2.0	4.0	V dc
Gate current	3411	Bias condition C, V_{GS} = +20 V; V_{DS} = 0	I_{GSSF1}		100		100	nA dc
Gate current	3411	Bias condition C, V_{GS} = -20 V; V_{DS} = 0	I_{GSSR1}		-100		-100	nA dc
Drain current	3413	Bias condition C, V_{GS} = 0 V_{DS} = 80 percent of rated V_{DS} (preirradiation)	I_{DSS}		10		10	µA dc
Static drain to source on-state voltage	3405	V_{GS} = 12 V; I_D = I_{D2} condition A, pulsed (see 4.5.1)	$V_{DS(on)}$					
2N7587U3, 2N7587U3C					0.855		0.855	V dc
2N7589U3, 2N7589U3C					1.104		1.104	V dc
2N7591U3, 2N7591U3C					1.340		1.340	V dc
2N7593U3, 2N7593U3C					1.638		1.638	V dc
Forward voltage source drain diode	4011	Bias condition C, V_{GS} = 0; I_D = I_{D1}	V_{SD}		1.2		1.2	V dc

1/ For sampling plan, see MIL-PRF-19500.
2/ Group D qualification may be performed prior to lot formation. Wafers qualified to these group D QCI requirements may be used for any other specification sheets utilizing the same die design.
3/ At the manufacturer's option, group D samples need not be subjected to the screening tests, and may be assembled in its qualified package or in any qualified package that the manufacturer has data to correlate the performance to the designated package.
4/ Separate samples shall be pulled for each bias.

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TABLE III. Group E inspection (all quality levels) for qualification or re-qualification only.

Inspection	MIL-STD-750		Sample plan
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling	1051	-55°C to +150°C, 500 cycles	
Hermetic seal Fine leak Gross leak	1071	 Test conditions G or H Test conditions C or D	
Electrical measurements		See table I, subgroup 2 herein.	
Subgroup 2 1/			45 devices c = 0
Steady-state gate bias	1042	Condition B, 1,000 hours.	
Electrical measurements		See table I, subgroup 2 herein.	
Steady-state reverse bias	1042	Condition A, 1,000 hours.	
Electrical measurements		See table I, subgroup 2 herein.	
Subgroup 3			n = 45, c = 0
Switching time test	3472	$I_D = I_{D1}$, V_{GS}= 4.5 Vdc, R_G = 24Ω, V_{DD} = 50 percent rated V_{DS}	
2N7587U3, 2N7587U3C		Typical measurements: $t_{d(on)}$ = 9.3 ns; t_r = 1.8 ns; $t_{d(off)}$ = 23.0 ns; t_f = 8.1 ns	
2N7589U3, 2N7589U3C		Typical measurements: $t_{d(on)}$ = 8.4 ns; t_r = 1.9 ns; $t_{d(off)}$ = 26.4 ns; t_f = 6.3 ns	
2N7591U3, 2N7591U3C		Typical measurements: $t_{d(on)}$ = 9.0 ns; t_r = 2.0 ns; $t_{d(off)}$ = 24.9 ns; t_f = 5.4 ns	
2N7593U3, 2N7593U3C		Typical measurements: $t_{d(on)}$ = 11.8 ns; t_r = 20.4 ns; $t_{d(off)}$ = 25.0 ns; t_f = 18.2 ns	
Subgroup 4			Sample size N/A
Thermal impedance curves		See MIL-PRF-19500.	
Subgroup 5			3 devices c = 0
Barometric pressure 2N7593U3, 2N7593U3C only	1001	To 70,000 feet	
Subgroup 6			3 devices
ESD	1020		
Subgroup 10			22 devices c = 0
Commutating diode for safe operating area test procedure for measuring dv/dt during reverse recovery of power MOSFET transistors or insulated gate bipolar transistors	3476		

See footnotes at end of table.

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TABLE III. Group E inspection (all quality levels) for qualification or re-qualification only - Continued.

Inspection	MIL-STD-750		Sample plan
	Method	Conditions	
Subgroup 11			3 devices
SEE 2/ 3/ 4/	1080	See figure 5	
Electrical measurements 5/		I_{GSSF1}, I_{GSSR1}, and I_{DSS1} in accordance with table I, subgroup 2	
SEE irradiation		Fluence = 3E5 ±20 percent ions/cm^2 Flux = 2E3 to 2E4 ions/cm^2/sec, temperature = 25 ±5 °C	
2N7587U3, 2N7587U3C		LET = 37.46 MeV-cm^2/mg, range = 36.03 microns, energy = 278.5 MeV In-situ bias conditions: V_{DS} = 100 V and V_{GS} = -19 V; V_{DS} = 40 V and V_{GS} = -20 V	
2N7589U3, 2N7589U3C		LET = 39 MeV-cm^2/mg, range = 39 microns, energy = 312 MeV In-situ bias conditions: V_{DS} = 150 V and V_{GS} = -20 V	
2N7591U3, 2N7591U3C		LET = 43 MeV-cm^2/mg, range = 205 microns, energy = 2441 MeV, In-situ bias conditions: V_{DS} = 200 V and V_{GS} = -10 V; V_{DS} = 190 V and V_{GS} = -15 V	
2N7593U3, 2N7593U3C		LET = 43 MeV-cm^2/mg, range = 112 microns, energy = 1217 MeV In-situ bias conditions: V_{DS} = 250 V and V_{GS} = -15 V ; V_{DS} = 100 V and V_{GS} = -17 V; V_{DS} = 50 V & V_{GS} = -20 V	
2N7587U3, 2N7587U3C		LET = 59.72 MeV-cm^2/mg, range = 30.97 microns, energy = 320 MeV, In-situ bias conditions: V_{DS} = 200 V and V_{GS} = -10 V; V_{DS} = 190 V and V_{GS} = -15 V	
2N7589U3, 2N7589U3C		LET = 59 MeV-cm^2/mg, range = 66 microns, energy = 825 MeV In-situ bias conditions: V_{DS} = 150 V and V_{GS} = -9 V; V_{DS} = 140 V and V_{GS} = -10 V; V_{DS} = 50 V and V_{GS} = -11 V; V_{DS} = 40 V and V_{GS} = -15 V	
2N7591U3, 2N7591U3C		In-situ bias conditions: V_{DS} = 200 V and V_{GS} = -15 V; V_{DS} = 190 V and V_{GS} = -20 V	
2N7593U3, 2N7593U3C		In-situ bias conditions: V_{DS} = 250 V and V_{GS} = -10 V; V_{DS} = 50 V and V_{GS} = -15 V	
2N7587U3, 2N7587U3C		LET = 81.44 MeV-cm^2/mg, range = 27.53 microns, energy = 333 MeV In-situ bias conditions: V_{DS} = 100 V and V_{GS} = -5 V	
2N7589U3, 2N7589U3C		LET = 90 MeV-cm^2/mg, range = 80 microns, energy = 1480 MeV In-situ bias conditions: V_{DS} = 50 V and V_{GS} = -5 V; V_{DS} = 30 V and V_{GS} = -10 V	
2N7591U3, 2N7591U3C		In-situ bias conditions: V_{DS} = 170 V and V_{GS} = -5 V	
2N7593U3, 2N7593U3C		In-situ bias conditions: V_{DS} = 75 V and V_{GS} = -5 V	
Electrical measurements 2/		I_{GSSF1}, I_{GSSR1}, and I_{DSS1} in accordance with table I, subgroup 2	

See footnotes at end of table.

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TABLE III. Group E inspection (all quality levels) for qualification or re-qualification only - Continued.

1/ A separate sample for each test shall be pulled.
2/ Group E qualification of SEE testing may be performed prior to lot formation. Qualification may be extended to other specification sheets utilizing the same structurally identical die design.
3/ Device qualification to a higher level linear energy transfer (LET) is sufficient to qualify all lower level LETs.
4/ The sampling plan applies to each bias condition.
5/ Examine I_{GSSF1}, I_{GSSR1}, and I_{DSS1} before and following SEE irradiation to determine acceptability for each bias condition. Other test conditions in accordance with table I, subgroup 2, may be performed at the manufacturer's option.

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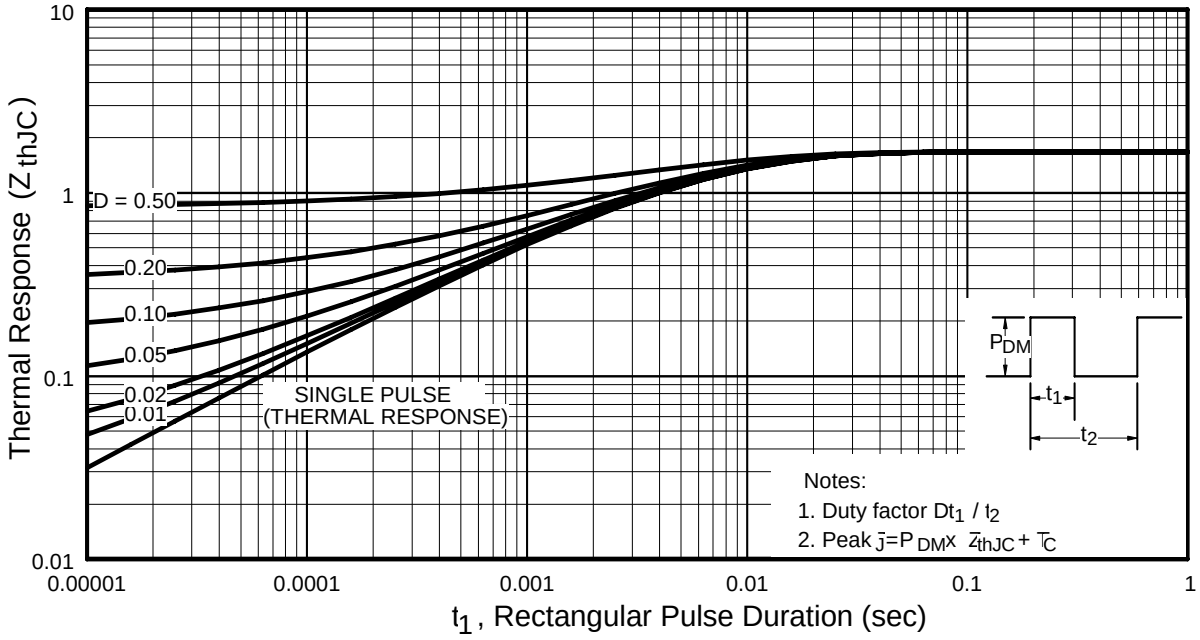
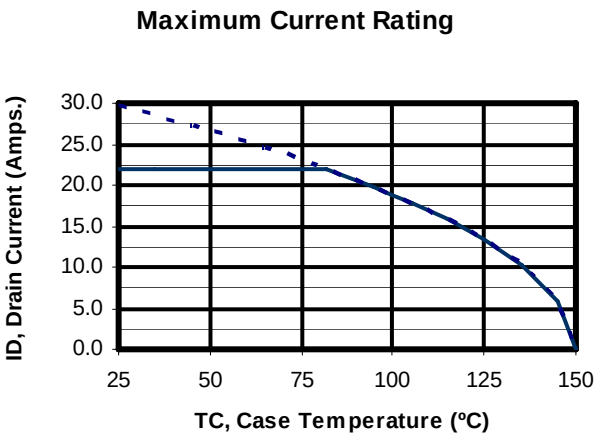
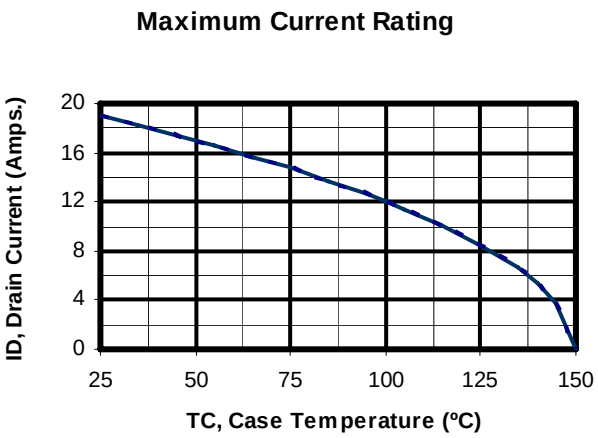


FIGURE 2. Thermal response curve.

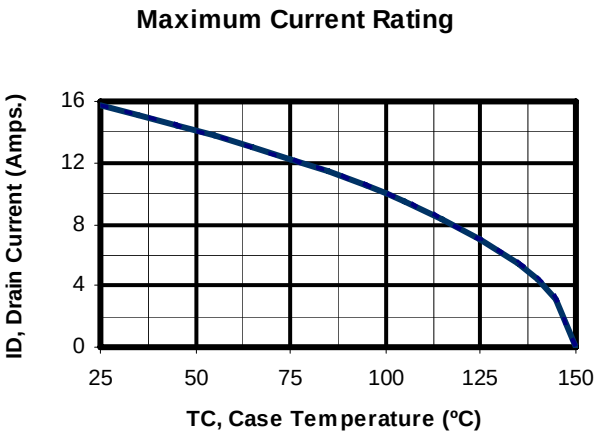
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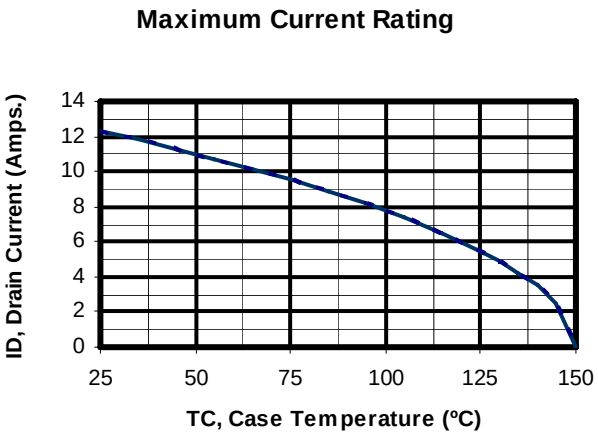
2N7587U3, 2N7587U3C



2N7589U3, 2N7589U3C



2N7591U3, 2N7591U3C

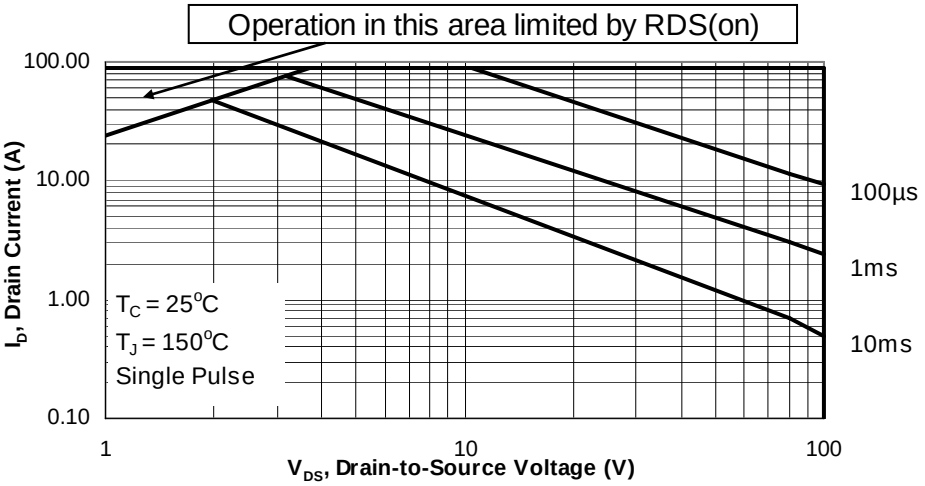


2N7593U3, 2N7593U3C

FIGURE 3. Maximum drain current versus case temperature graphs.

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2N7587U3, 2N7587U3C



2N7589U3, 2N7589U3C

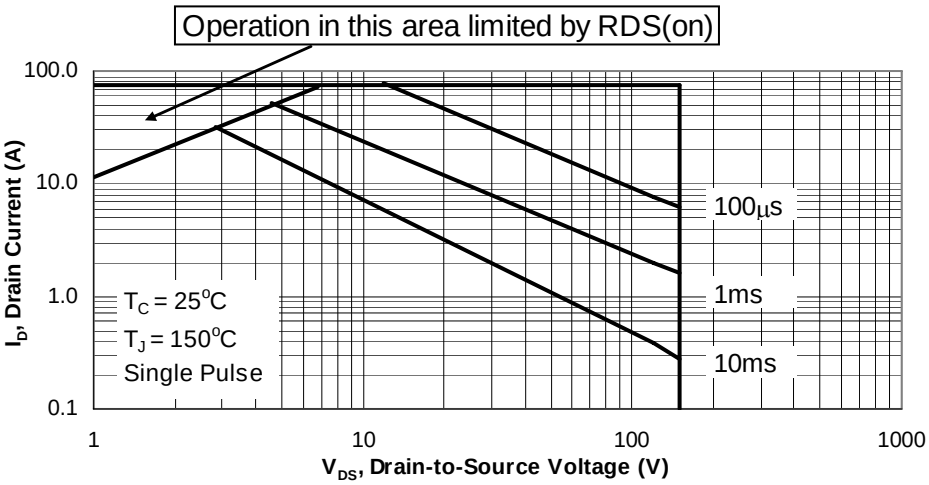
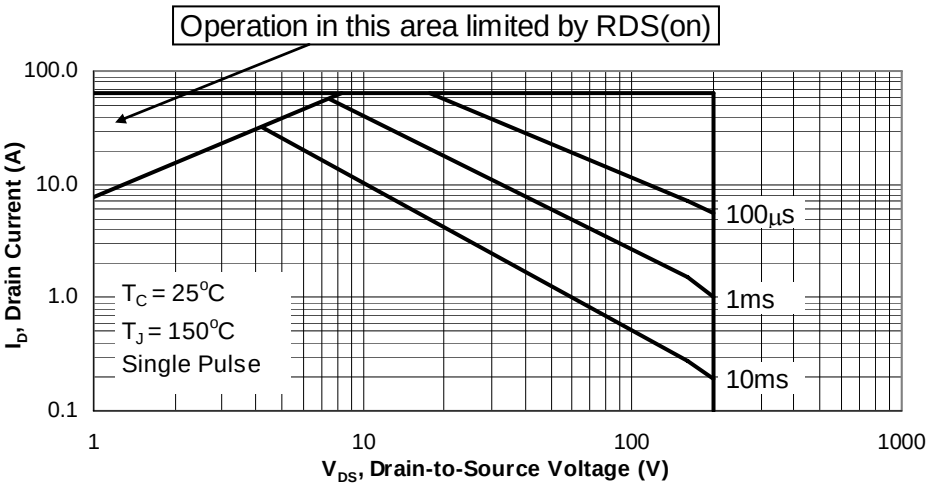


FIGURE 4. Safe operating area graph.

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2N7591U3, 2N7591U3C



2N7593U3, 2N7593U3C

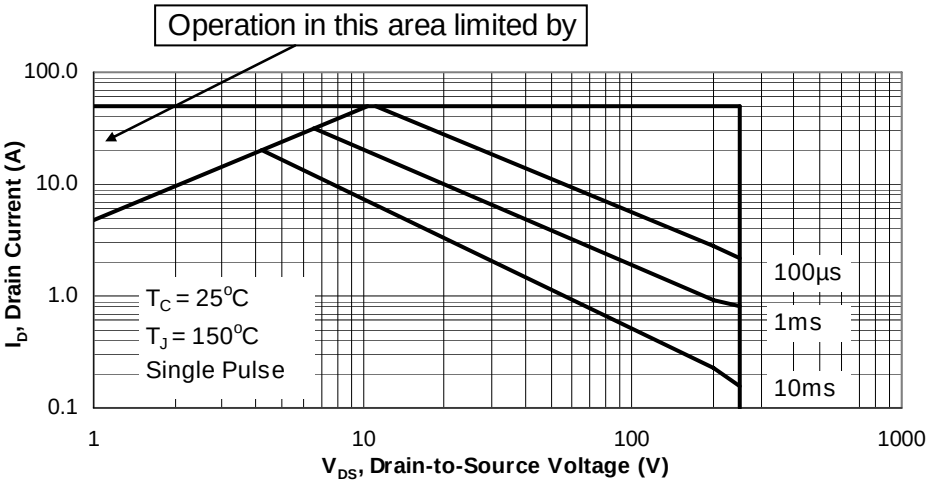


FIGURE 4. Safe operating area graph - Continued.

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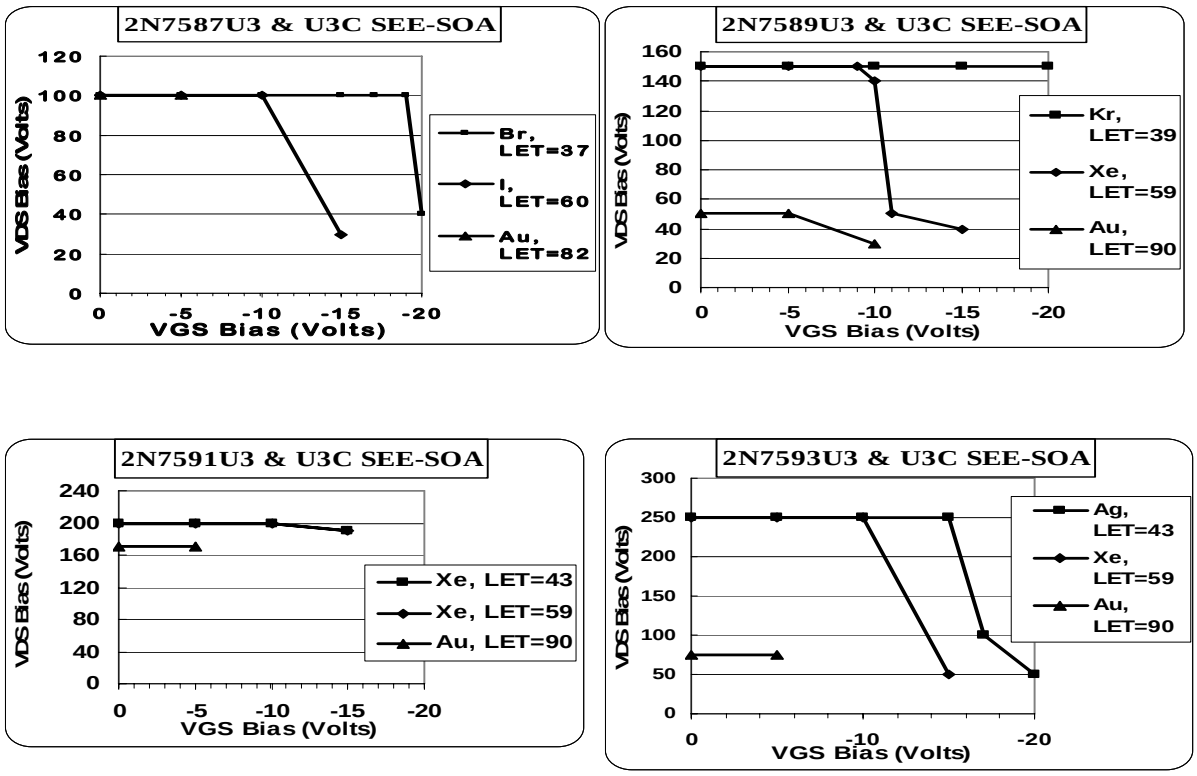


FIGURE 5. Typical SEE safe operating area graph.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. Product assurance level and type designator.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus, ATTN: DSCC/VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at http://assist.daps.dla.mil .

6.4 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's Part or Identifying Number (PIN) (without JAN and RHA prefix). This information in no way implies that manufacturer's PINs are substitutable for the military PIN.

Preferred types military PIN	Commercial PIN
2N7587U3 2N7589U3 2N7591U3 2N7593U3	IRHNJ67130 IRHNJ67134 IRHNJ67230 IRHNJ67234
2N7587U3C 2N7589U3C 2N7591U3C 2N7593U3C	IRHNJC67130 IRHNJC67134 IRHNJC67230 IRHNJC67234

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Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Preparing activity:
DLA - CC

(Project 5961-2008-041)

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